

Function Operations

Date _____ Period _____

Perform the indicated operation.

1) $f(x) = 3x + 4$
 $g(x) = -2x^2 - 4$
 Find $(f + g)(8)$

2) $h(n) = -n^3 - 2n$
 $g(n) = -2n - 1$
 Find $\left(\frac{h}{g}\right)(-4)$

3) $f(t) = 2t - 3$
 $g(t) = t^3 + t$
 Find $(f \cdot g)(0)$

4) $g(n) = -n + 5$
 $f(n) = n^2 - 1$
 Find $(g \circ f)(6)$

5) $g(x) = x + 2$
 $f(x) = x^3 - 2x$
 Find $(g \cdot f)(-4)$

6) $g(x) = x^2 + 2$
 $h(x) = 3x - 2$
 Find $(g + h)(-3)$

7) $h(x) = -x + 5$
 $g(x) = -3x - 2$
 Find $\left(\frac{h}{g}\right)(x)$

8) $g(x) = 4x - 2$
 $h(x) = x^2 - 5x$
 Find $g(x) - h(x)$

9) $f(x) = 2x + 5$
 $g(x) = 2x + 3$
 Find $(f + g)(x)$

10) $h(t) = 2t - 2$
 $g(t) = 4t + 4$
 Find $(h \cdot g)(t)$

11) $g(t) = t^2 - 2$
 $f(t) = 4t + 4$
 Find $g(t) \div f(t)$

12) $h(n) = 2n + 1$
 $g(n) = n - 2$
 Find $h(n) - 5g(n)$

13) $g(n) = 3n + 1$
 $h(n) = 2n - 3$
 Find $(-4g + 5h)(-2n)$

14) $h(n) = 3n - 1$
 $g(n) = 4n - 2$
 Find $(h \circ g)(4 + n)$

15) $g(t) = 4t + 4$
 $f(t) = t^2 + 2t$
 Find $(g - f)\left(\frac{t}{2}\right)$

16) $g(t) = 4t - 3$
 $f(t) = t^3 - 2$
 Find $(g + f)(-t)$

Find f and g so that $h(x) = (f \circ g)(x)$. Neither function may be the identity function $f(x) = x$.

17) $h(x) = \frac{5}{x^2} + 1$

18) $h(x) = (\sqrt{x} + 1)^2$

19) $h(x) = \sqrt{5x + 1} + 1$

20) $h(x) = 3^{\sqrt{x} + 1}$

21) $h(x) = (\sqrt{x} + 3)^2$

22) $h(x) = \frac{4}{x^2} + 2$